A Cross-Sectional Asset Pricing Test of Model Validity

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Abstract

An out-of-sample cross-sectional regression coefficient test of alpha in asset pricing models is investigated. The test is straightforward, easy to implement, applicable to linear and nonlinear models, and supplements in-sample GRS alpha tests of average pricing errors. Using U.S. stock returns, consistent with model misspecification, we find that the CAPM as well as prominent multifactor models have significant alpha coefficients in cross-sectional regressions. However, a recent model dubbed the ZCAPM is well specified with insignificant alpha coefficients. We conclude that the cross-sectional distribution of alphas across test assets contains useful information about model validity.

Keywords: alpha, asset pricing, cross-sectional tests, pricing error, ZCAPM
JEL: G12, C20

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1 Introduction

Jensen's (1968) alpha (denoted α) measures the abnormal return of a security equal to the difference between actual and predicted returns estimated by an empirical asset pricing model. Using alpha, he proposed that the CAPM of Sharpe (1964), Lintner (1965), and Mossin (1966) could be tested. In the case that $\alpha_i = 0$, an empirical model is considered to be well specified due to no pricing error.[1] Unfortunately, early studies of the CAPM found that alpha coefficients well exceeded zero.[2]

In a series of papers, Fama and French (1992, 1993, 1995, 1996a, 1996b, 1998) proposed that the market model should be augmented with size and value factors. These zero-investment portfolios are based on long/short portfolios of stocks sorted on firm characteristics (viz., market capitalization and book-to-market equity ratios to form size and value factors, respectively). They found that: "... for stock portfolios we examine, the intercepts from three-factor regressions ... are close to 0. Thus, a market factor and our proxies for size and book-to-market equity seem to do a good job explaining the cross-section of average stock returns."(Fama and French, 1993, p. 5) They observed that the market factor accounts for stock returns above the one-month Treasury bill rate, whereas the size and value factors explain differences in cross-sectional stock returns. Extending their analyses, Fama and French (1996a) found that the three-factor model accounted for a number of firm-characteristic anomalies, including size, earnings/price, cash flow/price, book-to-market equity, and past sales growth, but not anomalies related to momentum. Subsequently, Carhart (1997) augmented the three-factor model with a momentum factor defined as long past winner stocks and short

[1] See Merton (1973), Shanken (1987), Fama and French (1993), and others.

[2] See Black, Jensen, and Scholes (1972), Miller and Scholes (1972), Fama and MacBeth (1973), and others. Relative to a benchmark asset pricing model, positive (negative) alphas imply underpricing (overpricing) of assets.

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past loser stocks.[3] Moreover, augmenting their three-factor model, Fama and French (2015, 2018) later added profitability, capital investment, and momentum factors to obtain five- and six-factor models.

To comparatively evaluate the empirical validity of asset pricing models, given a sample of test assets, researchers often use the Gibbons, Ross, and Shanken (GRS) (1989) in-sample test for the joint zero equality of α_is in an estimation window.[4] Applicable to linear models, rejection of the null hypothesis indicates that a model is well specified. If average alpha is zero, there is no excess return to the factor risk premiums. The success of multifactors in reducing Jensen's alpha as well as significance in commonly-applied Fama and MacBeth (1973) cross-sectional tests led to a proliferation of new factors by researchers. Cochrane (2011) has referred to the growing population of factors as a *factor zoo*. To address this problem, Harvey, Liu, and Zhu (2016) and Chordia, Goyal, and Saretto (2020) examined over 300 multifactors and found that most are false discoveries. They recommended that factor loadings surpass a Fama-MacBeth cross-sectional t-statistic threshold of 3 to be considered significant. Thus, elimination of α_i intercepts in test assets is not a sufficient condition to accept a factor in particular and validate a factor model in general.

A recent paper by Hollstein and Prokopzuk (2022) examined a number of potential pitfalls in the GRS test, including look-ahead bias from using information that investors do not have, inappropriate constant-beta assumptions, rejection of models due to disparities between historical average factor returns and unobservable risk premia, and nonoptimal statistical properties.[5] For these reasons, they recommended standard Fama-MacBeth cross-sectional tests to evaluate different equity pricing models. In comprehensive tests, they documented that the Fama and French five-factor model fails cross-sectional tests due to relatively small

[3]See also studies by Jegadeesh and Titman (1993), Grundy and Martin (2001), Ehsani and Linnainmaa (2022), and citations therein on the momentum factor in asset pricing models.

[4]For example, see Cochrane (2005), Fama and French (2012), De Moor, Dhaene, and Sercu (2015), Fama and French (2015), Hou, Xue, Zhang (2015), Vorkink (2015), Stambaugh and Yuan (2017), Fama and French (2018), Fama and French (2020), Lettau and Pelger (2020), Giglio, Liao, and Xiu (2021), Hollstein and Prokopczuk (2022), and others.

[5]For further discussion of these econometric issues, see Barahona (2021), Lettau and Ludvigson (2001), Shanken (1987), and Bekaert and De Santis (2021), respectively, and citations therein.

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estimated factor risk premia and often insignificant factor loadings.

Contributing to cross-sectional tests of asset pricing models, we investigate a novel out-of-sample, Fama-MacBeth cross-sectional test of alphas. Our empirical test incorporates estimated first stage time-series regression alpha coefficients (or α_is) for a model as an independent variable in second stage out-of-sample cross-sectional regression analyses. If the alpha coefficient (denoted η) is significant in the cross-sectional regression for a given set of test assets, we infer that the model is misspecified due to pricing information contained in the cross-sectional distribution of alphas. Our test is straightforward, easy to implement, and supplements in-sample GRS tests of models. In contrast to the in-sample time-series regression GRS test, which focuses on average alphas across test assets, our out-of-sample cross-sectional regression test takes into account the dispersion of the sampling distribution of alphas. Also, unlike the GRS test that is appropriate for linear models, our cross-sectional test can be applied to both linear and nonlinear models. To our knowledge, no previous statistical tests in the asset pricing literature focus on the cross-sectional dispersion of alphas.

To demonstrate our out-of-sample cross-sectional tests of alphas, empirical tests are conducted using U.S. stock returns for the period July 1963 to December 2021. Prominent asset pricing models are estimated using time-series regressions, including the CAPM market model, Fama and French three-factor model, Carhart four-factor model, and Fama and French five- and six-factor models. Standard monthly rolling Fama and MacBeth cross-sectional regressions based on out-of-sample (one-month-ahead) excess returns of the test assets are estimated for each model. We find that estimated η coefficients associated with α_is are very significant for the asset pricing models under study in the cross section of average stock returns. Indeed, t-statistics for estimated η coefficients are very high in the range of 5 to 9, which suggests a strong relationship between alphas and one-month-ahead returns of test assets. These outsized t-statistics suggest model misspecification due to pricing error arising from either missing factors or non-factor related information contained in the cross-sectional distribution of alphas. Interestingly, after augmenting the CAPM and multifactor models with

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α_i coefficients, goodness-of-fit as estimated by cross-sectional regression R^2 values are very high near a perfect fit of 100 percent on an out-of-sample basis. We infer that valuable pricing information is embedded in the cross-sectional distribution of alphas. In this regard, for 25 size and book-to-market equity portfolios, while in-sample GRS tests of the three-factor model fail to reject the null hypothesis of jointly zero alphas, we find that alphas are significantly priced in our out-of-sample cross-sectional test. Since these two tests can yield different results, a more comprehensive approach to model validation is recommended that includes both tests.

A number of robustness checks corroborate our findings using only CAPM alphas, different test asset portfolios, individual stocks, and a recent asset pricing model dubbed that the ZCAPM by Kolari, Liu, and Huang (2021). The latter time-series regression model estimates beta risk related to average market returns and zeta risk associated with cross-sectional market return dispersion. Unlike the CAPM and multifactor models' test results, cross-sectional tests of the ZCAPM do not detect significant alphas for stock portfolios and individual stocks. These results suggests that the ZCAPM is well specified. Also, the insignificance of alphas in cross-sectional regression tests of the ZCAPM implies that our consistent finding of very significant alphas in other models is not a trivial mechanical result. Impressively, in standard out-of-sample cross-sectional regression tests without including alpha coefficients, the ZCAPM consistently outperforms other asset pricing models at times by large margins. Thus, the ZCAPM appears to dominate other well-known models.

In general, our empirical findings document that cross-sectional information in the distribution of alphas across test assets is potentially useful for validation of asset pricing models. Whereas the GRS test takes into account the equality of in-sample mean alphas for test assets, our out-of-sample cross-sectional test focuses on the relationship between the sampling distribution or dispersion of alphas and the cross section of average stock returns. For these reasons, we recommend that both in-sample time-series regression GRS tests and out-of-sample cross-sectional regression tests should be conducted by researchers to more

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comprehensively evaluate the validity of asset pricing models.

The next section reviews our methodology. Section 3 discusses the empirical results, implications to model validity, and robustness checks. The last section concludes.

2 Methodology

We propose a novel cross-sectional regression test for potential misspecification in asset pricing models. First, following the two-step Fama and MacBeth (1973) procedure, we estimate in the first stage a time-series OLS regression for each asset pricing model. We use both five years of monthly returns and one year of daily returns to estimate models. Since the empirical results are almost identical, to conserve space and avoid redundancy, we report the results based on one year of daily returns in forthcoming analyses.[6] For the ith test asset using daily returns from July 1963 to June 1964, we estimate:

$$R_{it} - R_{ft} = \alpha_i + \sum_{k=1}^{K} b_{ik} F_{kt} + e_{it}, \tag{1}$$

where $R_{it} - R_{ft}$ is the realized excess return on the ith test asset for day t, R_{ft} is the riskless rate proxied by the Treasury bill rate, α_i is the intercept term or Jensen's alpha, F_{kt} are asset pricing factors for $k = 1, \ldots, K$ factors, b_{ik} are corresponding K beta risk loadings for the ith portfolio, $i = 1, \ldots, N$ correspond to the number of portfolios, $t = 1, \ldots, T$ is the estimation period, and $e_{it} \sim$ iid $(0, \sigma_i^2)$. As already mentioned, it is standard practice to infer that, in the case of a well-specified asset pricing model, wherein all risk factors F_{kt} are taken into account, no pricing error exists such that intercept $\alpha_i = 0$. Consequently, if we specify only one factor in a model, but additional factors exist, the sum of the latter missing factor(s)

[6]Results using five years of daily returns are available upon request from the authors. While most researchers use monthly returns to estimate models, some authors employ daily returns (e.g., see Bali, Engle, and Tang, 2017). Additionally, in later robustness checks, we test a recent asset pricing model dubbed the ZCAPM by Kolari, Liu, and Huang (2021) for which daily returns are recommended by the authors for its empirical estimation.

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returns will be absorbed into pricing error term α_i as follows:

$$\alpha_i = \alpha_{0i} + \sum_{k=1}^{M} b_{ik} E(F_{kt}), \tag{2}$$

where the first component α_{0i} is associated with non-factor related information (e.g., unexplained anomalies), and the second component is generated by the missing factors $E(F_{kt})$ for $k = 1, \ldots, M$ (i.e., the total number of factors is $K + M$). Thus, α_i coefficients capture the total excess return for the ith test asset not attributable to factors in a model.

Second, we estimate a Fama-MacBeth cross-sectional regression in the next out-of-sample month. Numerous authors have recommended out-of-sample over in-sample tests of asset pricing models. Simin (2008), Ferson, Nallareddy, and Xie (2013), Barillas and Shanken (2018), Fama and French 2018), and others have observed that out-of-sample tests avoid data snooping issues and other econometric problems in explaining stock returns. Martin and Nagel (2021) have argued that, because investors are faced with many possible predictors of equilibrium price changes, out-of-sample tests are favored over in-sample tests.[7] Specifically, the following ordinary least squares (OLS) cross-sectional regression is estimated in July 1964:

$$R_{iT+1} - R_{fT+1} = R_z + \lambda_1 \hat{b}_{i1} + \lambda_2 \hat{b}_{i2} + \ldots + \lambda_K \hat{b}_{iK} + u_{iT+1}, i = 1, \ldots, N, \tag{3}$$

where $R_{iT+1} - R_{fT+1}$ is the realized excess return on the ith portfolio in out-of-sample (one-month-ahead) month $T + 1$, R_z in the intercept[8], $\hat{b}_{ik}, k = 1, \ldots, K$ are K beta risk loadings estimated for N portfolios in the estimation period $t = 1, \ldots, T$, λ_k are corresponding estimated market prices of beta risk loadings, and $u_{iT+1} \sim$ iid $(0,\sigma_i^2)$. Cochrane (2005, pp.

[7]Related to these issues, see also Lo and MacKinlay (1990) and Tian (2021) on problems of data snooping and model mining.

[8]According to Fama and French (2020), intercept R_z is a level return that equals the common return to all test assets not captured by the explanatory variables in the cross-sectional regression (i.e., the excess return on a standard portfolio of N test assets with weights that sum to one and zero out each explanatory variable). In the present context, denoting the intercept as R_z in cross-sectional regression (2) avoids confusion with the intercept or α_i coefficient in time-series regression (1).

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250-251) has noted that, in the Fama-MacBeth two stage procedure, the t-statistics associated with λ_ks are corrected for cross-sectional correlation of residual errors and therefore are equivalent to Shanken (1992) corrected OLS standard errors. Moreover, as observed by Ferson, Sarkissian, and Simon (1999), Back, Kapadia, Ostdiek (2013, 2015), and Ferson (2019), λ_ks are market prices of risk that represent long/short mimicking portfolios of stocks' factor loadings.

The above two regression equations are rolled forward one month at a time and repeated until the last cross-sectional regression in December 2021. Estimated monthly series of cross-sectional λ_k parameters are generated for July 1964 to December 2021 using daily returns (i.e., 690 λ_k estimates). These λ_k estimates are averaged over time, and t-statistics are computed to test the significance of their mean value $\bar{\lambda}_k$. Regarding the statistical properties of the Fama-MacBeth procedure, Shanken (1992) showed that maximum likelihood and two-pass cross-sectional estimators are asymptotically equivalent under standard joint normal assumptions about *iid* error terms. Under this condition, MacKinlay and Richardson (1991) found that the maximum likelihood (ML) method and the general method of moments (GMM) are asymptotically equivalent.[9] While GMM is believed to be useful if returns are not jointly normal and have conditional heteroscedasticity, Shanken and Zhou (2007) demonstrated by means of simulation analyses that residual return heteroscedasticity conditional on factors only marginally affects cross-sectional tests of asset pricing models. Also, they compared the cross-sectional regression risk premia estimates of the Fama and French three-factor model and found that size and value risk premia were close to sample means across different estimation methods, including OLS, weighted least squares (WLS), generalized least squares (GLS), GMM, and ML estimation.[10] Consistent with these findings, Giglio and Xiu (2021)

[9]According to Cochrane (2005, p. 251), the Fama and MacBeth estimate of the standard error is numerically the same as Shanken (1992) corrected standard errors.

[10]Relevant to this discussion, Shanken and Zhou (2007) recommended further simulation analyses of different cross-sectional estimation methods for a large number of test asset portfolios as employed in the present study. Such simulations are beyond the scope of the present paper. See also Jagannathan, Skoulakis, and Wang (2010) for an excellent discussion of econometric methods used in estimating linear beta pricing models.

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commented that WLS and GLS do not improve the asymptotic efficiency of OLS estimates to a first order; hence, as in the present study, they used OLS cross-sectional regression estimations.

Third, and last, departing from previous literature, we incorporate estimates of Jensen α_i parameters from each respective asset pricing model into out-of-sample cross-sectional regression (3) as follows:

$$R_{iT+1} - R_{fT+1} = R_z + \eta\hat{\alpha}_i + \lambda_1\hat{b}_{i1} + \lambda_2\hat{b}_{i2} + \ldots + \lambda_K\hat{b}_{iK} + u_{iT+1}, i = 1, \ldots, N, \tag{4}$$

where η measures the sensitivity of one-month-ahead excess returns of test assets to alpha pricing errors, and other notation is as before. The same two-step rolling Fama-MacBeth procedure is employed. Because the cross-sectional regressions employ one-month-ahead excess returns as the dependent variable, but the time-series regression model is based on daily returns, we multiply α_i by 21 to rescale it to a monthly basis. If η is not significantly different from zero, we infer that asset excess returns are not sensitive to alpha pricing error and, therefore, the model is well specified. However, if $\eta > 0$ due to a significant t-statistic, we infer that the model is misspecified due to the existence of non-factor related information, one or more missing factors, or other more significant factors.[11] Similar to the GRS test, cross-sectional Fama-MacBeth regressions assume a large number of test assets N and time periods T.

In equation (4), α_i has different units of measurement than b_{ik} in equation (1), which means that η has different units of measurement than λ_k. Time-series regression coefficient b_{ik} is dimensionless as it is measured without units (e.g., length, time, rate of return, etc.) Beta loading b_{ik} is recognized as a sensitivity coefficient reflecting the change in excess stock returns per unit change in factor returns. By contrast, intercept coefficient α_i has units of measurement in terms of returns and therefore is not a sensitivity coefficient. Analogously,

[11]It is possible that alpha is negative and significant in the case of a profitable trading strategy that shorts the respective factor.

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regression coefficient λ_k in equation (4) is a factor mimicking portfolio return in percent terms, whereas regression coefficient η is a dimensionless sensitivity coefficient reflecting the change in excess stock returns per unit change in α_i returns. In this regard, readers may incorrectly suppose that η is a factor mimicking portfolio return like λ_k in equation (4). To clarify matters, a mathematical proof in the Appendix confirms that sensitivity coefficient η measures the proportionate contribution of alpha pricing errors to the out-of-sample excess returns of a test asset.[12] Hence, we infer that η is not an estimate of the long/short factor mimicking portfolio return of high-alpha stocks' returns minus low-alpha stocks' returns. Just as the quantity $\lambda_k \hat{b}_{ik}$ in cross-sectional regression equation (4) captures the out-of-sample risk premium in return terms associated with the kth factor loading, the quantity $\eta\hat{\alpha}_i$ captures the out-of-sample risk premium in return terms attributable to missing factors and other non-factor related returns.

Lastly, we employ five prominent asset pricing models in our cross-sectional tests:

- CAPM market model with CRSP value-weighted index minus the Treasury bill rate as the market factor (M);
- Fama and French (1992, 1993) three-factor model (FF3) based on augmenting the market model with a size factor (small minus large firms' stock returns, or SMB) and value factor (high book-to- market equity minus low book-to-market equity firms' stock returns, or HML);
- Carhart (1997) four-factor model (CARHART) based on augmenting the three-factor model with a momentum factor (firms with high past return stock returns minus low past stock returns, or MOM);
- Fama and French (2015) five-factor model (FF5) based on augmenting their three-factor

[12]This interpretation can be readily verified by observing that, if we express returns in raw form rather than in terms of percent, the magnitude of η is unchanged, whereas the magnitudes of λ_k estimates would decrease, as they represent the factor mimicking return in raw return form now. As further confirmation, we estimated time-series regression (1) using one year of daily returns and rolled forward one day at time instead of one month at a time. The magnitudes of η are little changed, but the magnitudes of intercept R_z as well as market prices of risk λ_ks are lower by a factor of about 21. These results verify that η is a sensitivity coefficient, rather than a market price of risk.

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model with a profitability factor (robust operating profitability minus weak operating profitability returns, RMW) and capital investment factor (conservative investment minus aggressive investment returns, or CMA);[13] and

- Fama and French (2018) six-factor model (FF6) based on augmenting five-factor model with a momentum factor.

Based on the earlier discussed procedures, we estimate cross-sectional regressions (3) and (4) for these models without and with α_i coefficient augmentation, respectively.

2.1 Data

Daily rate of returns series for U.S. stocks are downloaded from Kenneth French's website for the period July 1963 to December 2021. The following seven portfolio sets are combined into one composite portfolio comprised of 175 stock portfolios: 25 size (market capitalization) and book-to-market equity portfolios, 25 size and profitability portfolios, 25 size and capital investment portfolios, 25 size and momentum portfolios, 25 book-to-market equity and profitability portfolios, 25 book-to-market equity and capital investment portfolios, and 25 profitability and capital investment portfolios. These portfolios are representative of often-studied firm-characteristic anomalies in stock returns. Additionally, we combine the 175 anomaly portfolios with 47 industry portfolios[14] to obtain a total of 222 stock portfolios. We added industry portfolios due to recommendations by Lewellen, Nagel, and Shanken (2010), Daniel and Titman (2021), and others to include exogenous industry test assets in asset pricing tests. In this regard, Ferson, Sarkissian, and Simin (1999) have cautioned against the potential hazards of using attribute-sorted portfolios to test asset pricing models. As such, we provide tests for both composite sets of 175 and 222 portfolios. In robustness checks discussed in forthcoming Section 3.5, we utilize both constituent portfolios in the 222 portfolios as well as 1,000 individual stocks with the largest market capitalization at the end of each year in our

[13]This model is similar to the q-factor model of Hou, Xue, and Zhang (2015).

[14]The French data website makes available historical daily returns for 48 and 49 industry portfolios, but only 47 industries contained daily returns in our full sample period.

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sample period.[15] While other test assets could be employed, our purpose is to demonstrate the proposed cross-sectional test rather than make exhaustive tests of asset pricing models.

Descriptive statistics for the asset pricing factors are provided in Table 1. As shown there, the market factor ($R_m - R_f$) and momentum factor (MOM) have noticeably higher mean daily returns (in percent) and standard deviations of daily returns than the multifactors SMB, HML, RMW, and CMA.

3 Empirical Results

The plan of this section is as follows. Descriptive statistics for alphas across different asset pricing models and test assets document their mean values and standard deviations. Graphical analyses of the in-sample alphas over time illustrate the time-series behavior of alphas for models. Out-of-sample cross-sectional regression alpha tests provide the main findings of our analyses. Subsequently, we discuss related implications to model validity tests. Lastly, we conduct a number of robustness checks that confirm our main findings.

3.1 Alpha Estimation for Models

Table 2 reports the descriptive statistics for alphas (αs) estimated using regression equation (1) for the asset pricing models. In each one year estimation window, the mean alpha based on α_is for the 222 test asset portfolios is computed, or $\bar{\alpha}$, as well as the standard deviation of α_is. As the regression model is estimated using daily returns for one year and then rolled forward one month at a time throughout our sample period, a time series of 690 $\bar{\alpha}$s and standard deviations of α_is are constructed. Again, daily α_i are multiplied by 21 to scale them to monthly percent returns. The overall mean $\bar{\alpha}$ and overall standard deviation of α_is are

[15] Black, Jensen, and Scholes (1972) grouped stocks into portfolios to avoid an error-in-the-variables problem in estimating beta loadings in asset pricing models. Portfolios reduce this problem to a second order of importance. However, portfolios may not accurately capture valuable information about the validity of an asset pricing model. For this reason, as discussed in forthcoming Subsection 3.4.3, some researchers have recommended further tests using individual stocks.

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computed for all monthly time series values in addition to minimum and maximum $\bar{\alpha}$s.[16] As shown in Table 2, the overall mean $\bar{\alpha}$ as well as minimum and maximum $\bar{\alpha}$s are much larger for the CAPM compared to those for the multifactor models. By contrast, while the standard deviation of α_is for the CAPM is greater than those for the multifactor models, the disparities are less than those for mean $\bar{\alpha}$s.

3.2 Graphical Analyses of Alphas Over Time

For the comprehensive set of 222 test asset portfolios, Figure 1 graphically displays the time series of mean daily alpha estimates, or $\bar{\alpha}$, in Table 2 for the models. For each portfolio, rolling forward one month at a time generates 690 one-year regressions, wherein 690 $\bar{\alpha}$ estimates are computed. The mean alpha in each one year period is computed across the 222 portfolios. These in-sample graphical analyses are repeated for each asset pricing model.

In Panel A of Figure 1, we see that $\bar{\alpha}$ estimates across different models have similar intertemporal patterns. The CAPM has noticeably larger $\bar{\alpha}$ magnitudes and volatility over time compared to the multifactor models. By excluding the CAPM, Panel B in Figure 1 more clearly shows the close co-movement of $\bar{\alpha}$s among different multifactor models. Indeed, the FF3, CARHART, and FF5 models have high correlation coefficients (ρ) of $\bar{\alpha}$s with those for the FF6 model equal to 0.73, 0.88, and 0.88, respectively; hence, these multifactor models have very similar time series of $\bar{\alpha}$s. By comparison, the CAPM has correlation coefficients (ρ) of $\bar{\alpha}$s with those for the FF3, CARHART, FF5, and FF6 models equal to 0.26, 0.14, 0.33, and 0.42, respectively. These relatively low correlations suggest that CAPM alphas contain pricing errors not present in multifactor models' alphas. Importantly, Panel B of Figure 1 demonstrates that, as multifactors are incrementally added to the models, the magnitudes and volatility of $\bar{\alpha}$ over time are unchanged for the most part.

In Figure 2, we graph the cross-sectional standard deviation of alphas (α_is) for the CAPM and FF6 model. Here the standard deviation of α_is is computed across the 222 portfolios (not

[16]To conserve space, we do not report similar descriptive statistics for the 175 sorted portfolios as well as 47 industry portfolios, which are comparable in most respects.

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across time) in each estimation window. The one-year regression is rolled forward one month at a time to the end of 2021. By casual inspection, the magnitudes of standard deviations are visibly higher for the CAPM compared to the FF6 model. Using one year of daily returns to estimate time-series regressions, the correlation coefficient (ρ) of standard deviations of α_is for the CAPM and FF6 model is quite high at 0.81. Even so, due to the differences in the levels of standard deviations, it is obvious that CAPM alphas contain cross-sectional pricing errors not explained by the FF6 model.

3.3 Out-of-Sample Cross-Sectional Alpha Tests

Using daily returns in the prior year to estimate the first stage time-series regression for each model, we report the empirical results for out-of-sample cross-sectional regressions rolling one month at a time for 175 and 222 portfolios in Tables 3 and 4, respectively. Models augmented with α_i coefficients are appended with an A (e.g., the augmented CAPM is denoted CAPM-A).

Using 175 test asset portfolios in Table 3, the estimated sensitivity coefficients denoted η associated with alpha pricing errors in cross-sectional regression equation (4) are highly significant with t-statistics ranging from 6.02 to 7.25. This level of significance is exceptional in comparison to cross-sectional estimates of market prices of risk λ_k (associated with the kth factor loadings) reported in extant asset pricing literature and exceeds the recommended 3.0 t-statistic threshold by Harvey, Liu, and Zhu (2016) and Chordia, Goyal, and Saretto (2020). Thus, we infer that significant pricing information exists in the cross-sectional distribution of alphas.

When factor loadings are significantly priced for different models, the magnitudes of their market prices of risk and significance levels are little changed by adding α_i coefficients to the base models without these coefficients. For example, we find significant market prices of risk for profitability factor loadings, or λ_{RMW}, with monthly magnitudes (t-statistics) equal to 0.18 percent (2.47), 0.23 percent (2.05), 0.17 percent (2.34), and 0.22 percent (2.90) in the FF5-A,

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FF5, FF6-A, and FF6 models, respectively. In augmented and non-augmented cross-sectional regression results, the estimated market prices of size and capital investment factor loadings, or λ_{SML} and λ_{CMA}, range from 0.18 percent to 0.24 percent and 0.13 percent to 0.15 percent, respectively, with t-statistics in the range from 1.67 to 2.12 (i.e., between 5 percent and 10 percent significance levels). Value factor factor loadings are normally insignificant, but momentum loadings reach the 5 percent significant level in the CARHART and FF6 models. Consistent with previous empirical literature but contrary to CAPM theory, market factor loadings (or market beta loadings for portfolios) are not significant in any model and have negative coefficient signs in most cases. Since market prices of risk of different factor loadings are not materially changed by the inclusion of α_i coefficients, we infer that risk premiums impounded in alphas are not correlated with those of multifactors under study.

It is noteworthy that the α_i coefficient materially boosts the goodness-of-fit of the cross-sectional regressions.[17] For example, the CAPM's goodness-of-fit increases dramatically from an estimated R^2 equal to 7 percent to 96 percent for the CAPM-A. The latter near perfect goodness-of-fit is extraordinary on an out-of-sample basis. Clearly, the CAPM is misspecified and missing factors are implied by our α_i test results in the cross-sectional regression.

The comparative R^2 results for multifactor models are as follows: Fama and French three-factor model increases from 29 percent (FF3) to 98 percent (FF3-A), Carhart four-factor model increases from 52 percent (CARHART) to 98 percent (CARHART-A), Fama and French five-factor model increases from 55 percent (FF5) to 98 percent (FF5-A), and Fama and French six-factor model increases from 71 percent (FF6) to 98 percent (FF6-A). Thus, in terms of goodness-of-fit, even the strongest six-factor model is noticeably boosted by augmentation with α_i coefficients. These results suggest that multifactor models suffer from material misspecification related to pricing information contained in the sampling distribution of alphas. Notice that, as multifactors have been added to the models, their goodness-of-fit

[17]Following standard methods in Jagannathan and Wang (1996) and Lettau and Ludvigson (2001, footnote 17, p. 1254), we estimate goodness-of-fit as the R^2 value from a single equation cross-sectional regression (viz., the time-series averages of one-month-ahead excess returns for all test assets are regressed on their time-series average fitted or predicted excess returns).

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progressively improves but missing factors continue to be suggested by significant estimated η coefficients. The FF6 model results with a highly significant η coefficient at 0.17 (t-statistic = 6.76) imply that either more factors or more significant factors will be found by researchers.

Upon incorporating 47 industry portfolios to obtain 222 test asset portfolios, the results in Table 4 further support alpha pricing errors as a significant explanatory variable in the asset pricing models under study. As shown there, compared to the results for 175 portfolios, now the t-statistics associated with sensitivity coefficient η are even higher in the range of 6.68 to 7.73. Other results are similar to those in Table 3; however, estimated R^2 values noticeably decrease compared to Table 3 due to the addition of exogenous industry portfolios to the test assets. Even so, as before, goodness-of-fit of the models is considerably enhanced by α_i coefficients. Again the FF6 model has a highly significant η coefficient at 0.19 (t-statistic = 7.51), which suggests the existence of other factors or more significant factors.

It is worth mentioning that the magnitudes of estimated η coefficients in Tables 3 and 4 are economically substantial in the range of 0.17 to 0.27. In this respect, assume that $\eta = 1$. In this case, 100 percent of missing factor returns are contained in the out-of-sample (one-month-ahead) excess returns of test asset portfolios. Applying this interpretation, our results indicate that 17 percent to 27 percent of the magnitudes of α_i intercepts (or pricing error) contribute to out-of-sample test asset portfolio excess returns.

3.4 Implications to Model Validity

Summarizing our findings, based on out-of-sample cross-sectional regression tests, six factors in the FF6 model are unable to eliminate the significance of sensitivity coefficient η, which consistently has very high t-statistics in the range of 6.76 to 7.51 using the composite 175 and 222 test asset portfolios, respectively. The very high significance of η coefficients in these tests occurs despite relatively small in-sample alphas across test assets in the FF6 model. Consistent with many previous studies (see footnote 4), as factors are added to the models, average alphas become progressively closer to zero.

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Using the 222 test asset portfolios, we conduct GRS tests for both monthly returns spanning the entire sample period as well as one year of daily returns that roll forward one month at a time until December 2021 (to allow time-varying factor loadings in models). For the latter tests, average αs, average GRS test statistics, and average p-values are computed from monthly rolling regressions in the sample period for each model. Our results for the entire sample period reject the joint zero equality of α_is. The F-statistic is 2.88 for the CAPM, which gradually decreases as factors are added to 2.45 for the FF6 model with p-value below 0.001. Hollstein and Prokopczuk (2022, p. 5) estimated the FF5 model and obtained a similar GRS test statistic of 2.48 for the period January 1963 to December 2018 (i.e., we estimate this test statistic to be 2.55 for the FF5 model in our sample period). Regarding tests using the one month rolling approach, out of 690 rolling months, about 28 percent (15 percent) of GRS test statistics are significant at the 5 percent level or lower for the CAPM (FF6 model). We infer that multifactors diminish average pricing errors but do not eliminate them. These findings are consistent with our cross-sectional tests of alpha sensitivity coefficients, or η, which indicate misspecification in extant asset pricing models.[18]

We earlier quoted Fama and French (1993), who observed that their three-factor model had near zero alphas or intercept terms in empirical tests of 25 size and book-to-market equity sorted portfolios. Hence, they inferred that the model explained the returns of these anomaly portfolios and therefore was well specified. Replicating their three-factor model tests with these portfolios and monthly returns in the sample period July 1963 to December 1991, we conduct GRS tests. The in-sample GRS F-statistic is 1.49 ($p = 0.115$), such that α_is are jointly zero at the 10 percent confidence level consistent with Fama and French. Upon re-estimating the GRS test in our sample period July 1963 to December 2021, we obtain $F = 3.75$, which indicates that α_is are not jointly zero. Interestingly, using their sample period, our out-of-sample cross-sectional test of alpha sensitivity coefficients yields $\eta = 0.37$ with $t = 2.89$ significant at the 1 percent level. In our sample period, we estimate $\eta = 0.38$

[18]We should note that, whether or not alphas are jointly zero per GRS tests, it is possible that alphas are not significant in cross-sectional tests of the associated η coefficient.

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with $t = 4.08$. We infer that significant mispricing is suggested in both sample periods. Thus, even though in-sample α_is are jointly zero in Fama and French's (1993) sample period, model misspecification is indicated by our out-of-sample cross-sectional regression α_i test. In a series of later papers, Fama and French (2015, 2018), 2020) added new multifactors to their earlier three-factor model to improve its performance, which is in line with our cross-sectional regression tests of η coefficients associated with alpha mispricing errors.

An important implication of our findings is that in-sample GRS tests should be supplemented with out-of-sample cross-sectional tests to further assess pricing errors and model misspecification. Even if in-sample GRS tests suggest that α_is are reduced or near zero as factors are added to models, pricing errors captured by the cross-sectional distribution of alpha may well contain significant pricing information as indicated by out-of-sample cross-sectional regression tests. If α_is are significant in cross-sectional regression tests, missing factors or more significant asset pricing factors exist in all likelihood.

3.5 Robustness Checks

This section reviews the empirical findings for a number of robustness checks.

3.4.1 CAPM Alphas

We re-run the cross-sectional regression analyses using only α_i coefficients from the CAPM. That is, only α_is from the time-series CAPM market model are used in cross-sectional tests of the FF3-A, CARHART-A, FF5-A, and FF6-A models. Interestingly, the estimates and significance of the alpha sensitivity coefficients ηs are almost identical to those in Tables 3 and 4 using α_i coefficients from each respective time-series regression model.[19] We infer that, while multifactor models reduce the mean value of α_is relative to the market model, they do not materially change the cross-sectional distribution of α_is for test asset portfolios and therefore cross-sectional regression alpha test results.

To further explore this inference, we conduct correlation analyses of the α_is of different

[19]These results are available from the authors upon request.

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asset pricing models. In each one year estimation period, the correlations of the time-series CAPM market model α_is and other time-series models' α_is are computed. The average for 690 one-month-rolling one year periods (from rolling forward one month at a time in the sample period) generate the following average correlation coefficients ($\bar{\rho}$): CAPM α_is and FF3 α_is ($\bar{\rho} = 0.75$), CAPM α_is and CARHART α_is ($\bar{\rho} = 0.73$), CAPM α_is and FF5 α_is ($\bar{\rho} = 0.47$), and CAPM α_is and FF6 α_is ($\bar{\rho} = 0.50$). These results indicate that the cross-sectional distributions of α_is across different models are correlated over time and therefore contain similar information about pricing errors.

3.4.2 Constituent Portfolios

We repeat the cross-sectional analyses for each of the constituent 25 portfolios in the composite 175 composite dataset as well as 47 industry portfolios. In unrecorded results to conserve space, while the results are similar to the larger composite portfolios, with very significant estimated η coefficients, some interesting patterns in the goodness-of-fit emerge. Excluding industry portfolios and using daily returns, CAPM-A consistently produces very high goodness-of-fit with estimated R^2 values in the range of 93 percent to 98 percent for different test assets of 25 portfolios. The CAPM-A does well based on industry portfolios with an R^2 value equal to 90 percent. Other mulitifactor models with α_i coefficient augmentation have R^2 values that are similar to or less than those for CAPM-A. We infer that CAPM-A does a good job of explaining out-of-sample portfolio returns due to capturing pricing error information embedded in the cross-sectional distribution of α_i coefficients.

3.4.3 Individual Stocks

According to Ang, Liu, and Schwarz (2020), more efficient asset pricing tests are possible with individual stocks compared to portfolios, especially with respect to the second stage cross-sectional regression analysis. In view of this potential advantage, we repeat the cross-sectional regression analyses for the top 1,000 individual stocks by market capitalization in the CRSP common stock database. These stocks account for over 90 percent of total market capitalization. To construct these daily return series, on the last trading day in each one

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year model estimation period, we select the 1,000 individual stocks with the largest market capitalization and then collect their daily returns for that year.

For sample individual stocks, mean alphas and their standard deviations are similar for the CAPM and FF6 model (as well as other models). In Figure 3, we graph the mean alphas (Panel A) and standard deviations of alpha (Panel B) for rolling one year CAPM and FF6 model regressions over time. The magnitudes of mean alphas for individual stocks in Panel A are much more similar to one another than in Figure 1 based on the 222 composite portfolios. Also, standard deviations of alphas over time in Panel B are close to one another for the CAPM and FF6 model, which again exceeds their similarity in earlier Figure 2 for the 222 portfolios. At least for individual stocks, the CAPM appears to do as well as the multifactor models in terms of pricing error. Importantly, in out-of-sample cross-sectional regression tests shown in Table 5, again alpha sensitivity coefficients (ηs) are very significant with t-statistics ranging from 4.64 o 4.84 for different models containing α_i coefficients. Only profitability factor loadings (i.e., λ_{RMW}) are significant at the one percent level (i.e., all other factors' loadings are not significant at the 10 percent level). These results confirm significant misspecification in these asset pricing models.

3.4.4 ZCAPM Tests

Lastly, we extend our empirical tests to a recent asset pricing model proposed by Kolari, Liu, and Huang (2021) (KLH). The authors derived a special case of Black's (1972) zero-beta CAPM dubbed the ZCAPM.[20] The model assumes the same general equilibrium conditions as Black's now famous model. An alternative but equivalent form of the zero-beta ZCAPM, the ZCAPM has two readily estimable factors: (1) the mean market return (i.e., CRSP index return), and (2) their cross-sectional return dispersion (i.e., a nontradable factor). As cited by KLH, previous authors consider the return dispersion of stock returns in the market to be a macroeconomic state variable related to unemployment rates, economic uncertainty,

[20]See also Liu, Kolari, and Huang (2012), Liu, Kolari, and Huang (2012), Liu (2013), Liu, Kolari, and Huang (2020), Kolari, Liu, and Huang (2020), Kolari, Huang, Butt, and Liao (2022), Kolari, Huang, Liu, and Liao (2022, 2023, 2024), Kolari and Pynnonen (2023), and Kolari, Liu, and Pynnonen (2024).

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business cycles, macroeconomic shocks, etc.[21] Some studies have augmented the CAPM and multifactor models with a return dispersion factor, including Jiang (2010), Demirer and Jategaonkar (2013), and Garcia, Mantilla-Garcia, and Martellini (2014); however, these studies reported mixed results for the return dispersion factor.[22] Unlike these studies that rely upon OLS regression models, KLH reported strong cross-sectional test results for return dispersion based on an empirical ZCAPM with a latent dummy signal variable, which is estimated via the expectation-maximization (EM) algorithm.[23]

KLH make available Matlab, R, and Python programs for EM estimation of the empirical ZCAPM as well as two stage Fama-MacBeth regression tests on GitHub (https://Github.com/zcapm). In this study, we employ their R programs and daily returns for one year as in KLH to estimate the following empirical ZCAPM:

$$R_{it} - R_{ft} = b_{iM}(R_{Mt} - R_{ft}) + Z^*_{iM}\sigma_{Mt} + u^*_{it}, \quad t = 1, \cdots, T. \tag{5}$$

where σ_{Mt} is the value-weighted cross-sectional standard deviation of returns for all U.S. common stocks on day t (i.e., return dispersion), Z^*_{iM} measures sensitivity to daily return dispersion σ_{Mt} due to so-called *zeta risk*; and other notation is as before. The zeta risk coefficient Z^*_{iM} is computed by multiplying the regression coefficient Z_{iM} by the probability of a dummy hidden variable (viz., $+1, -1$) estimated by the EM algorithm (see KLH for

[21] See Loungani, Rush, and Tave (1990), Christie and Huang (1994), Duffee (2001), Connolly and Stivers (2003), Gomes, Kogan, and Zhang (2003), Zhang (2006), Adrian and Rosenberg (2008), Chichernea, Holder, and Petkevich (2015), Choudhry, Papadimitriou, and Shabi (2016), Demirer, Gupta, Zhihu, and Wing-Keung (2019), and others. Researhers have used macroeconomic factors to capture risks in the economy that explain cross-sectional expected returns, including Chen, Roll, and Ross (1986), Adrian, Etula, and Muir (2014), and others.

[22] It is important to point out that cross-sectional return dispersion is different from time-series volatility (e.g., the Chicago Board Options Exchange's (CBOE) Volatility Index, or VIX), which is a separate branch of the asset pricing literature. Numerous studies investigate time-series market volatility, including Ang, Hodrick, Xing, and Zhang (2006), Bekaert, Engstrom, and Ermolov (2023), Detzel, Duarte, Kamara, and Siegel (2024), and citations therein.

[23] See seminal work on the EM algorithm by Dempster, Laird, and Rubin (1977) in addition to studies by Jones and McLachlan (1990), McLachlan and Peel (2000), and McLachlan and Krishnan (2008), among others. For applications of the EM algorithm in finance studies, see Wu (1983), Kon (1984), Rudd (1991), Asquith, Jones, and Kieschnick (1998), McLachlan and Krishnan (2008), Harvey and Liu (2016), Chen, Cliff, and Zhao (2017), and citations therein. EM regression methods are widely used in the hard sciences.

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details).[24] As already mentioned, cross-sectional return dispersion σ_{Mt} is a nontradable factor measured in percent returns, which is computed as follows:

$$\sigma_{Mt} = \sqrt{\frac{N}{N-1}\sum_{i=1}^{N} w_{it-1}(R_{it} - R_{Mt})^2}, \tag{6}$$

where N is the total number of stocks in the CRSP index (excluding stocks with stock prices less than \$5 per share), w_{it-1} is the previous day's market value weight for the ith stock (i.e., market capitalization of the stock divided by the market capitalization of all N stocks), and other notation is as before.

Notice that there is no pricing error (i.e., $\alpha_i = 0$) in empirical ZCAPM equation (5).[25] Following Fama and French (2020, p. 1893), who estimated pricing error for a model without an intercept term using the average time-series residual error term, based on empirical ZCAPM relation (5), we compute an estimate of α_i as the average u_{it} over $t = 1, \cdots, T$ for the ith portfolio or individual stock, or $\bar{u}_{it}$. Using this estimate of α_i, we also compute the mean $\bar{\alpha}_i$ and standard deviation of α_is across the $i = 1, \cdots, N$ portfolios or stocks. As before for other models using one year of daily excess returns from July 1963 to June 1964, equation (5) is estimated for each of the composite 222 stock portfolios and then repeatedly estimated by rolling forward one month at a time ending at November 2021 (i.e., one month before the end of the sample period as December 2021 is the last one-month-ahead period for the rolling cross-sectional regressions). Also, we repeat these analyses for our sample of 1,000 individual stocks.

Focusing on the ZCAPM and Fama and French six-factor model (FF6), Figure 4 plots mean alphas ($\bar{\alpha}$) and their cross-sectional standard deviations, which are computed for the 222 portfolios using 690 one-month-rolling estimated regressions.[26] Panel A of Figure 4 shows

[24]The use of hidden or latent variables is well accepted in empirical asset pricing models. Numerous studies have utilized Principal component analysis (PCA) and factor analysis to create hidden factors in asset pricing models. See, for example, Roll and Ross (1980), Chen (1983), Lettau and Pelger (2020), and citations therein.

[25]Experiments by KLH found that alternative methods of estimating an α_i term did not improve in-sample data fitting with lower residual variance.

[26]The graphs in Figure 4 change little for the 175 portfolios and therefore are not reported.

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that $\bar{\alpha}$s are similar for these two models and fluctuate around zero; indeed, their overall means over 690 rolling estimates of $\bar{\alpha}$s are near zero at only 0.038 percent per month for the ZCAPM and 0.045 percent for the FF6 model. By contrast, as shown in Panel B of Figure 4, their cross-sectional standard deviations of $\bar{\alpha}$s differ markedly over time. The ZCAPM has an overall mean standard deviation of $\bar{\alpha}$s over 690 rolling estimates equal to 0.268 percent compared to 0.718 percent for the FF6 model. Hence, the cross-sectional distribution of alphas for the ZCAPM are noticeably lower that those for the FF6 model.

Because the ZCAPM is a nonlinear model containing a cross-sectional volatility factor, it is not possible to run in-sample GRS tests, which are appropriate for linear asset pricing models. Hence, another advantage of our out-of-sample cross-sectional alpha test is that it can be applied to both linear and nonlinear models. Turning to cross-sectional tests of the ZCAPM without and with alpha augmentation (ZCAPM-A), Table 6 summarizes the results for 175 firm and stock characteristic sorted portfolios (Panel A), 222 portfolios including 47 industries (Panel B), and the top 1,000 individual stocks by market capitalization (Panel C). For the 175 and 222 composite portfolios, alpha sensitivity coefficients ηs are positive but insignificant with t-statistics equal to 1.48 and 1.43, respectively. In sharp contrast to our earlier findings for the CAPM and multifactor models, this insignificance implies that the ZCAPM is well specified. Additionally, the insignificance of alphas in the ZCAPM implies that the repeated highly significant alphas for other models is not a trivial mechanical result.

A major reason for the insignificance of the η coefficient in ZCAPM-A is that zeta risk loadings are highly significant with λ_{RD} coefficients in the range of 0.42 percent to 0.44 percent (i.e., monthly zeta risk premiums) and very high t-statistics from 5.43 to 6.05. Notice that the latter t-statistics associated with zeta risk loadings are considerably higher than those for loadings associated with multifactors in earlier tables and well exceed the 3 threshold recommended by Harvey, Liu, and Zhu (2016) and Chordia, Goyal, and Saretto (2020).[27]

[27]KLH (2021, p. 167) reported a similarly high t-statistic of 5.42 for zeta risk loadings using U.S. stock returns in the analysis period January 1965 to December 2018 for 97 portfolios (comprised of 25 size-BM sorted, 25 beta-zeta sorted, and 47 industry portfolios). Corroborating these findings, in a recent working paper, Kolari, Huang, Liu, and Liao (2024) documented that zeta risk loadings are highly significant using

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Also, while estimated R^2 values increase somewhat in both sets of portfolios from the ZCAPM to ZCAPM-A with alpha coefficient augmentation from 89 percent to 94 percent in Panel A and from 82 percent to 91 percent in Panel B, the ZCAPM has markedly higher R^2 values than earlier reported for the five asset pricing models. Among the different models tested in Table 3, the FF6 model had the highest R^2 value at 71 percent, and in Table 4 the maximum R^2 was again the FF6 model at 54 percent. As noted earlier, the inclusion of 47 industry portfolios in Table 4 materially reduced the R^2 value of the FF6 model. Interestingly, the ZCAPM has goodness-of-fit more comparable to the augmented asset pricing models than the multifactor models themselves. It clearly dominates the other models in standard Fama-MacBeth cross-sectional tests.

The results for the top 1,000 individual stocks in Panel C of Table 6 are similar to the composite portfolios, except that now sensitivity coefficient η is positive and marginally significant at the 10 percent level with t-statistic equal to 1.72. Notice that the λ_{RD} coefficient for the market price of zeta risk loadings is again highly significant but its magnitude declines to 0.16 percent and 0.15 percent per month in the ZCAPM-A and ZCAPM with t-statistics 3.43 and 3.45, respectively. Nonetheless, assuming a 5 percent significance level threshold, we infer from these results that alpha pricing errors for sample individual stocks are largely explained by zeta risk premiums in the ZCAPM. Compared to the other models, recall from Table 5 that only profit factor loadings in the FF5 and FF6 models are statistically significant with a maximum t-statistic of 2.84. Thus, the ZCAPM again outperforms the other models.

In summary, the ZCAPM outperforms prominent asset pricing models in out-of-sample cross-sectional regression tests with and without α_i coefficients. Alpha sensitivity coefficients η for the ZCAPM are insignificant for test asset portfolios but nominally significant at the 10 percent level for individual stocks. We infer that the significance of alpha coefficients

large numbers of anomaly portfolios as test assets (i.e., 133 and 153 anomaly portfolios made available online as open source datasets by researchers). Also, they found that the ZCAPM did a good job of explaining out-of-sample average pricing errors (i.e., actual returns minus predicted returns) of anomaly portfolios. By contrast, the CAPM and multifactor models tested in the present paper performed poorly in their study. They concluded that the ZCAPM outperforms prominent asset pricing models and helps to explain a large cross section of anomaly portfolio returns.

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in cross-sectional regressions of the other asset pricing models is not due to a mechanical relation between time-series regression alpha returns and out-of-sample (one-month-ahead) portfolio returns in their respective cross-sectional regressions. The alpha coefficients become insignificant in the case of the ZCAPM, which implies that they are nontrivial in the other models. Also, our results suggest that the ZCAPM is well specified.

4 Conclusion

This paper investigated a novel out-of-sample Fama and MacBeth (1973) cross-sectional regression test of α_i intercepts in asset pricing models. If the alpha sensitivity coefficient (denoted η) is significant, we infer that pricing information is contained in the cross-sectional distribution of alphas. Straightforward and easy to implement, our test supplements in-sample GRS tests for the joint zero equality of α_is by providing an out-of-sample test of the cross-sectional dispersion of α_is. Also, it can be applied to both linear and nonlinear asset pricing models, rather than only linear models per the GRS test,

As shown for U.S. stock returns, out-of-sample statistical tests of α_i coefficients (denoted η) were consistently very significant for the CAPM as well as prominent multifactor models. We infer that these models are misspecified. A variety of robustness checks confirmed these results. In further empirical tests using a recent asset pricing model dubbed the ZCAPM, alpha sensitivity coefficients (η) were insignificant in cross-sectional regressions for different test assets. As such, the ZCAPM is well specified. Interestingly, even without the inclusion of α_i intercepts, standard out-of-sample cross-sectional regression tests of the ZCAPM consistently outperformed other asset pricing models at times by large margins. Thus, compared to prominent asset pricing models, the ZCAPM better explains the cross section of average stock returns in our tests.

An important implication confirmed by empirical tests is that, even if in-sample GRS tests indicate mean alphas jointly near or equal to zero for an asset pricing model, in the

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case that out-of-sample estimates of η are significantly greater than zero, valuable pricing information can be embedded in the cross-sectional sampling distribution of alphas of test assets. As such, we recommend that researchers conduct more comprehensive alpha tests of model validation, including both in-sample time-series regression GRS tests and out-of-sample cross-sectional regression tests. In this regard, further out-of-sample cross-sectional tests of alphas for different models and test assets are warranted. Lastly, in view of its superior asset pricing performance in cross-sectional tests, additional research on the ZCAPM is recommended.

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Table 1: Descriptive statistics for asset pricing factors in the sample period July 1963 to December 2021

This table reports descriptive statistics for asset pricing factors in the sample period July 1963 to December 2021. Results for average daily returns are shown in percent. Variables are defined as follows: R_f (riskless rate) = U.S. Treasury bill return, R_m (market index) = value-weighted CRSP common stock index return, *SMB* (size factor) = small minus big firms' stock returns, *HML* (value factor) = high B/M minus low B/M firms' stock returns, *MOM* (momentum factor) = high past return stocks minus low past return stocks, *RMW* (profitability factor) = robust operating profitability minus weak operating profitability returns, and *CMA* (capital investment factor) = conservative investment minus aggressive investment returns. Daily data series are downloaded from Kenneth French's website.

	Mean	Std dev	Min	Max
R_f	0.0174	0.0127	0.00	0.06
$R_m - R_f$	0.0284	1.0122	-17.44	11.35
SMB	0.0076	0.5404	-11.23	6.17
HML	0.0138	0.5617	-5.02	6.74
MOM	0.0280	0.7555	-14.38	7.12
RMW	0.0137	0.3849	-3.02	4.51
CMA	0.0128	0.3648	-5.89	2.53

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Table 2: Descriptive statistics for time-series regression estimates of alphas (αs) for different asset pricing models in the sample period July 1963 to November 2021

This table reports descriptive statistics for mean alphas ($\bar{\alpha}$) estimated using one year of daily returns in different asset pricing models for 222 test asset portfolios. The regression for each model is estimated from July 1963 to June 1964, and subsequent regressions roll forward one month at a time until November 2021. Mean αs are multiplied by 21 to scale them to monthly percent returns. Time-series regression equation (1) in the text is estimated for the following models: CAPM market model (CAPM), Fama and French three-factor model (FF3), Carhart four-factor model (CARHART), Fama and French five-factor model (FF5), and Fama and French six-factor model (FF6). The overall mean $\bar{\alpha}$ is the average for the time series of 690 $\bar{\alpha}$s from monthly rolling regressions. Overall std dev α_i is similarly computed using the month-by-month estimates of the standard deviation of α_is for the portfolios. Min and max $\bar{\alpha}$s are shown also.

Model	Overall mean $\bar{\alpha}$	Overall std dev α_i	Min $\bar{\alpha}$	Max $\bar{\alpha}$
CAPM	0.1119	0.2794	-1.18	1.65
FF3	0.0163	0.2091	-0.24	0.45
CARHART	0.0211	0.1986	-0.23	0.54
FF5	0.0435	0.1945	-0.23	0.49
FF6	0.0446	0.1932	-0.16	0.47

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Table 3: Cross-sectional regression results rolling one month at a time using daily returns in one year in time-series regressions for 175 test asset portfolios: July 1964 to December 2021

This table reports the cross-sectional regression results for different asset pricing models using daily returns. Daily excess return series for 175 U.S. stock portfolios downloaded from Kenneth French's website are employed as test assets (see text). First, using one year of daily returns from July 1963 to June 1964, we estimate time-series regression equation (1) in the text for the following asset pricing models: CAPM market model (CAPM), Fama and French three-factor model (FF3), Carhart four-factor model (CARHART), Fama and French five-factor model (FF5), and Fama and French six-factor model (FF6). Second, using out-of-sample (one-month-ahead) excess returns in July 1964, we run Fama and MacBeth (1973) cross-sectional regression equation (2) using the estimated risk loadings (coefficients) from the time-series regression for each model. This two-step process is rolled forward one month at a time to December 2021. Averages of estimated market prices of risk, or λ_k, for $1, \ldots, K$ factors (and associated t-statistics) are shown in the table. We estimate the following market prices (corresponding to respective factor loadings): λ_M (market factor loading), λ_{SMB} (size factor loading), λ_{HML} (value factor loading), λ_{MOM} (momentum factor loading), λ_{RMW} (profitability factor loading), and λ_{CMA} (capital investment factor loading). Third, departing from previous literature, each cross-sectional regression model (3) is re-run after augmentation with Jensen's α_i coefficients from its respective time-series regression model (1). We denote these models by appending the letter A (e.g., CAPM becomes CAPM-A). The sensitivity coefficient with respect to α_i coefficients is denoted η. Riskless U.S. Treasury bill rates, CRSP value-weighted returns, and factors' returns for *SMB*, *HML*, *MOM*, *RMW*, and *CMA* are downloaded from French's website.

Model	R_z	η	λ_M	λ_{SMB}	λ_{HML}	λ_{MOM}	λ_{RMW}	λ_{CMA}	R^2
CAPM-A	0.54	0.27	0.12						0.96
	(2.86)	(6.02)	(0.57)						
CAPM	0.78		-0.09						0.07
	(3.81)		(-0.43)						
FF3-A	0.56	0.20	0.03	0.18	0.15				0.98
	(3.56)	(6.83)	(0.18)	(1.67)	(1.48)				
FF3	0.69		-0.09	0.22	0.16				0.29
	(4.23)		(-0.49)	(1.92)	(1.52)				
CARHART-A	0.69	0.19	-0.10	0.20	0.16	0.15			0.98
	(4.52)	(7.25)	(-0.57)	(1.88)	(1.61)	(1.07)			
CARHART	0.82		-0.23	0.22	0.18	0.31			0.52
	(5.28)		(-1.28)	(1.93)	(1.73)	(2.03)			
FF5-A	0.47	0.18	0.13	0.20	0.13		0.18	0.15	0.98
	(2.94)	(6.50)	(0.74)	(1.87)	(1.26)		(2.47)	(2.04)	
FF5	0.60		0.01	0.24	0.15		0.23	0.14	0.55
	(3.63)		(0.05)	(2.12)	(1.38)		(2.95)	(1.97)	
FF6-A	0.71	0.17	-0.11	0.23	0.15	0.18	0.17	0.14	0.98
	(4.80)	(6.76)	(-0.67)	(2.12)	(1.44)	(1.23)	(2.34)	(1.91)	
FF6	0.84		-0.23	0.24	0.17	0.30	0.22	0.13	0.71
	(5.53)		(-1.38)	(2.06)	(1.56)	(1.94)	(2.90)	(1.90)	

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Table 4: Cross-sectional regression results rolling one month at a time using daily returns in one year in time-series regressions for 222 test asset portfolios: July 1964 to December 2021

This table reports the cross-sectional regression results for different asset pricing models using daily returns. Daily excess return series for 222 U.S. stock portfolios downloaded from Kenneth French's website are employed as test assets (see text). First, using one year of daily returns from July 1963 to June 1964, we estimate time-series regression equation (1) in the text for the following asset pricing models: CAPM market model (CAPM), Fama and French three-factor model (FF3), Carhart four-factor model (CARHART), Fama and French five-factor model (FF5), and Fama and French six-factor model (FF6). Second, using out-of-sample (one-month-ahead) excess returns in July 1964, we run Fama and MacBeth (1973) cross-sectional regression equation (2) using the estimated risk loadings (coefficients) from the time-series regression for each model. This two-step process is rolled forward one month at a time to December 2021. Averages of estimated market prices of risk, or λ_k, for $1, \ldots, K$ factors (and associated t-statistics) are shown in the table. We estimate the following market prices (corresponding to respective factor loadings): λ_M (market factor loading), λ_{SMB} (size factor loading), λ_{HML} (value factor loading), λ_{MOM} (momentum factor loading), λ_{RMW} (profitability factor loading), and λ_{CMA} (capital investment factor loading). Third, departing from previous literature, each cross-sectional regression model (3) is re-run after augmentation with Jensen's α_i coefficients from its respective time-series regression model (1). We denote these models by appending the letter A (e.g., CAPM becomes CAPM-A). The sensitivity coefficient with respect to α_i coefficients is denoted η. Riskless U.S. Treasury bill rates, CRSP value-weighted returns, and factors' returns for *SMB*, *HML*, *MOM*, *RMW*, and *CMA* are downloaded from French's website.

Model	Intercept	η	λ_M	λ_{SMB}	λ_{HML}	λ_{MOM}	λ_{RMW}	λ_{CMA}	R^2
CAPM-A	0.49	0.25	0.17						0.95
	(2.85)	(6.68)	(0.91)						
CAPM	0.70		-0.01						0.02
	(3.84)		(-0.06)						
FF3-A	0.47	0.21	0.12	0.20	0.12				0.97
	(3.29)	(7.28)	(0.65)	(1.93)	(1.25)				
FF3	0.60		-0.00	0.21	0.14				0.18
	(4.06)		(-0.01)	(1.84)	(1.36)				
CARHART-A	0.56	0.20	0.03	0.21	0.12	0.14			0.97
	(3.85)	(7.73)	(0.15)	(1.99)	(1.20)	(0.97)			
CARHART	0.66		-0.06	0.21	0.15	0.37			0.37
	(4.45)		(-0.36)	(1.82)	(1.43)	(2.37)			
FF5-A	0.39	0.20	0.21	0.21	0.10		0.17	0.15	0.97
	(2.61)	(7.16)	(1.16)	(2.04)	(1.01)		(2.29)	(2.05)	
FF5	0.50		0.11	0.23	0.12		0.20	0.15	0.38
	(3.32)		(0.59)	(2.02)	(1.18)		(2.58)	(2.11)	
FF6-A	0.60	0.19	0.00	0.23	0.12	0.16	0.17	0.12	0.97
	(4.10)	(7.51)	(0.00)	(2.12)	(1.16)	(1.14)	(2.26)	(1.70)	
FF6	0.69		-0.08	0.22	0.14	0.37	0.20	0.13	0.54
	(4.67)		(-0.49)	(1.95)	(1.37)	(2.36)	(2.52)	(1.80)	

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Table 5: Cross-sectional regression results rolling one month at a time using daily returns in one year in time-series regressions for the top 1,000 individual stocks by market capitalization: July 1964 to December 2021

This table reports the cross-sectional regression results for the top 1,000 individual stocks by market capitalization for different asset pricing models using daily returns. Daily excess return series for U.S. individual stocks downloaded from CRSP are employed as test assets (see text). First, using one year of daily returns from July 1963 to June 1964, we estimate time-series regression equation (1) in the text for the following asset pricing models: CAPM market model (CAPM), Fama and French three-factor model (FF3), Carhart four-factor model (CARHART), Fama and French five-factor model (FF5), and Fama and French six-factor model (FF6). Second, using out-of-sample (one-month-ahead) excess returns in July 1964, we run Fama and MacBeth (1973) cross-sectional regression equation (2) using the estimated risk loadings (coefficients) from the time-series regression for each model. This two-step process is rolled forward one month at a time to December 2021. Averages of estimated market prices of risk, or λ_k, for $1, \ldots, K$ factors (and associated t-statistics) are shown in the table. We estimate the following market prices (corresponding to respective factor loadings): λ_M (market factor loading), λ_{SMB} (size factor loading), λ_{HML} (value factor loading), λ_{MOM} (momentum factor loading), λ_{RMW} (profitability factor loading), and λ_{CMA} (capital investment factor loading). Third, departing from previous literature, each cross-sectional regression model (3) is re-run after augmentation with Jensen's α_i coefficients from its respective time-series regression model (1). We denote these models by appending the letter A (e.g., CAPM becomes CAPM-A). The sensitivity coefficient with respect to α_i coefficients is denoted η. Riskless U.S. Treasury bill rates, CRSP value-weighted returns, and factors' returns for *SMB*, *HML*, *MOM*, *RMW*, and *CMA* are downloaded from French's website.

Model	R_z	λ_α	λ_M	λ_{SMB}	λ_{HML}	λ_{MOM}	λ_{RMW}	λ_{CMA}
CAPM	0.57		0.10					
	(4.20)		(0.60)					
FF3-A	0.45	0.10	0.15	-0.00	0.14			
	(3.92)	(4.67)	(1.02)	(-0.02)	(1.52)			
FF3	0.47		0.15	0.05	0.09			
	(4.05)		(1.01)	(0.55)	(0.94)			
CARHART-A	0.45	0.09	0.14	0.01	0.10	0.01		
	(4.07)	(4.84)	(0.95)	(0.11)	(1.21)	(0.04)		
CARHART	0.50		0.12	0.06	0.05	0.09		
	(4.34)		(0.87)	(0.75)	(0.65)	(0.66)		
FF5-A	0.43	0.09	0.20	0.02	0.09		0.20	0.06
	(3.85)	(4.64)	(1.39)	(0.26)	(1.00)		(2.84)	(0.85)
FF5	0.45		0.20	0.07	0.04		0.18	0.06
	(4.04)		(1.40)	(0.84)	(0.42)		(2.64)	(0.94)
FF6-A	0.44	0.09	0.17	0.02	0.06	0.03	0.20	0.04
	(4.05)	(4.82)	(1.28)	(0.28)	(0.75)	(0.25)	(2.83)	(0.66)
FF6	0.48		0.16	0.07	0.02	0.11	0.17	0.04
	(4.36)		(1.18)	(0.86)	(0.22)	(0.84)	(2.62)	(0.66)

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Table 6: Cross-sectional regression results rolling one month at a time for the empirical ZCAPM using daily returns in one year for test asset portfolios: July 1964 to December 2021

This table reports the cross-sectional regression results for the empirical ZCAPM using daily returns. Daily excess return series for U.S. stocks for both 175 firm and stock characteristic sorted portfolios (Panel A) and 222 portfolios including the 175 portfolios plus 47 industry portfolios (Panel B) downloaded from Kenneth French's website are employed at test assets (see text). Results for the top 1,000 individual stocks by market capitalization at year-end of each one year estimation window. First, using one year of daily returns from July 1963 to June 1964, we estimate empirical ZCAPM equation (8) in the text by means of the expectation-maximization (EM) regression analysis. Second, using out-of-sample (one-month-ahead) excess returns in July 1964, we run Fama and MacBeth (1973) cross-sectional regression equation (2) using the estimated risk loadings (coefficients) from the time-series regression for each model. This two-step process is rolled forward one month at a time to December 2021. Averages of estimated market prices of risk, or λ_k, for $1, \ldots, K$ factors (and associated t-statistics) are shown in the table. We estimate the following market prices (corresponding to respective factor loadings): λ_M (market factor loading) and λ_{RD} (cross-sectional return dispersion factor loading). Third, the ZCAPM is re-run after augmentation with Jensen's α_i estimates (denoted ZCAPM-A), which are computed as the average of the residual error terms. The sensitivity coefficient with respect to α_i coefficients is denoted η. Daily riskless U.S. Treasury bill rates and CRSP value-weighted returns are downloaded from French's website. Daily cross-sectional return dispersion for the ZCAPM is computed as the cross-sectional value-weighted standard deviation of returns for all common stocks on CRSP. Because R^2 values are computed using a single equation cross-sectional regression for the entire analysis period, no estimated R^2 values are shown for individual stocks that can change in the sample of top 1,000 stocks over time.

Panel A. 175 portfolios

Model	Intercept	η	λ_M	λ_{RD}	R^2
ZCAPM-A	0.51	0.14	0.14	0.44	0.94
	(2.72)	(1.48)	(0.72)	(5.43)	
ZCAPM	0.56		0.10	0.44	0.89
	(2.95)		(0.49)	(5.47)	

Panel B. 222 portfolios

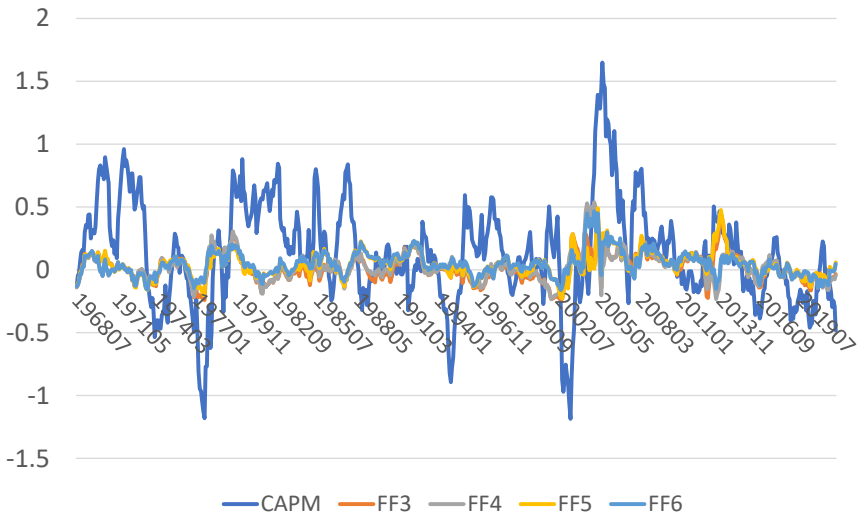
Model	Intercept	η	λ_M	λ_{RD}	R^2
ZCAPM-A	0.50	0.12	0.14	0.43	0.91
	(2.99)	(1.43)	(0.80)	(6.05)	
ZCAPM	0.53		0.14	0.42	0.82
	(3.09)		(0.73)	(6.03)	

Panel C. Top 1,000 individual stocks by market capitalization

Model	Intercept	η	λ_M	λ_{RD}	R^2
ZCAPM-A	0.53	0.07	0.09	0.16	na
	(4.22)	(1.72)	(0.55)	(3.43)	
ZCAPM	0.53		0.11	0.15	na
	(4.18)		(0.66)	(3.45)	

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Panel A. Mean alphas for all models



Panel B. Mean alphas for models excluding CAPM

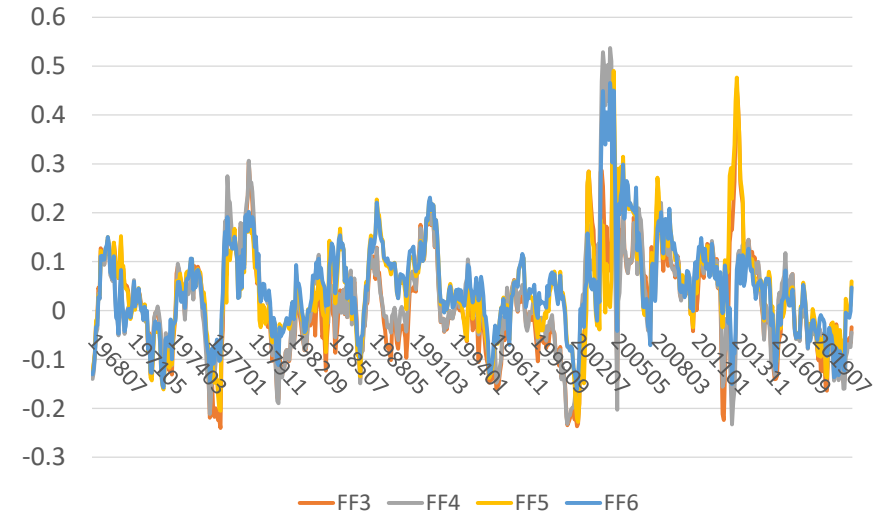


Figure 1: This graph shows one-month rolling mean alpha ($\bar{\alpha}$) values estimated from time-series regressions of different asset pricing models using one year of daily excess returns for 222 stock portfolios. Panel A includes all models (CAPM, FF3, CARHART, FF5, and FF6), and Panel B excludes the CAPM. Mean α_is are multiplied by 21 to scale them to monthly percent returns. The time-series regression for each model is initially estimated using daily returns from July 1963 to June 1964, and subsequent regressions are estimated by rolling forward one month at a time until November 2021 to obtain series of 690 values.

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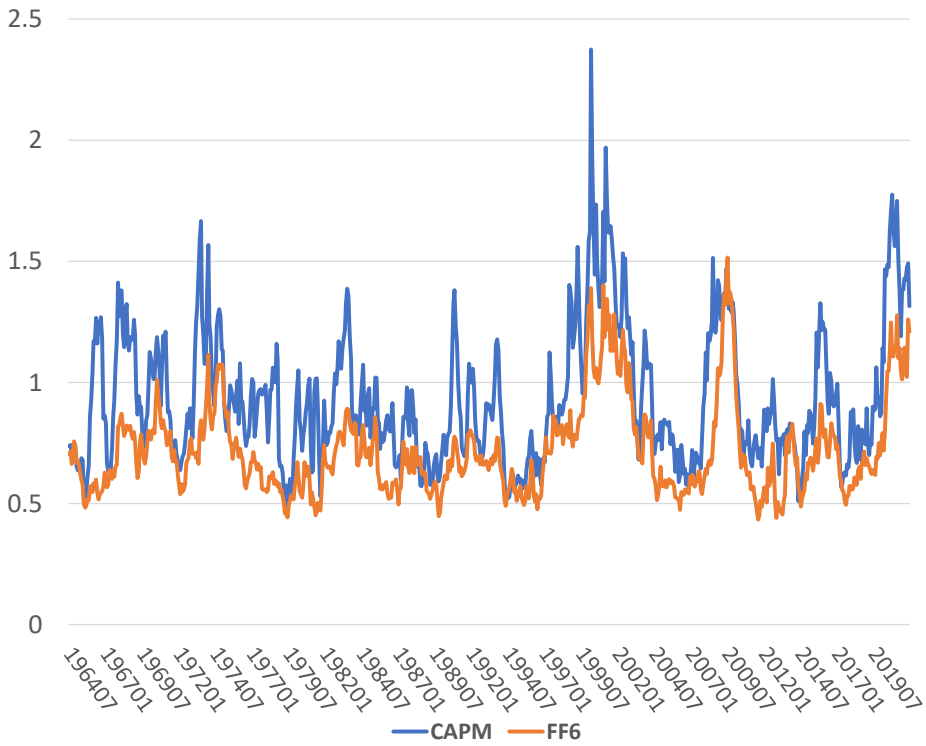


Figure 2: This graph shows the cross-sectional standard deviations of alpha (α_i) values from time-series regressions of the CAPM and FF6 model for 222 stock portfolios. Daily α_i values are multiplied by 21 to scale them to monthly percent returns, such that the standard deviation is in monthly percent terms. The time-series regression for each model is initially estimated using one year of daily returns from July 1963 to June 1964, and subsequent regressions are estimated by rolling forward one month at a time to the last month November 2021 to obtain series of 690 values.

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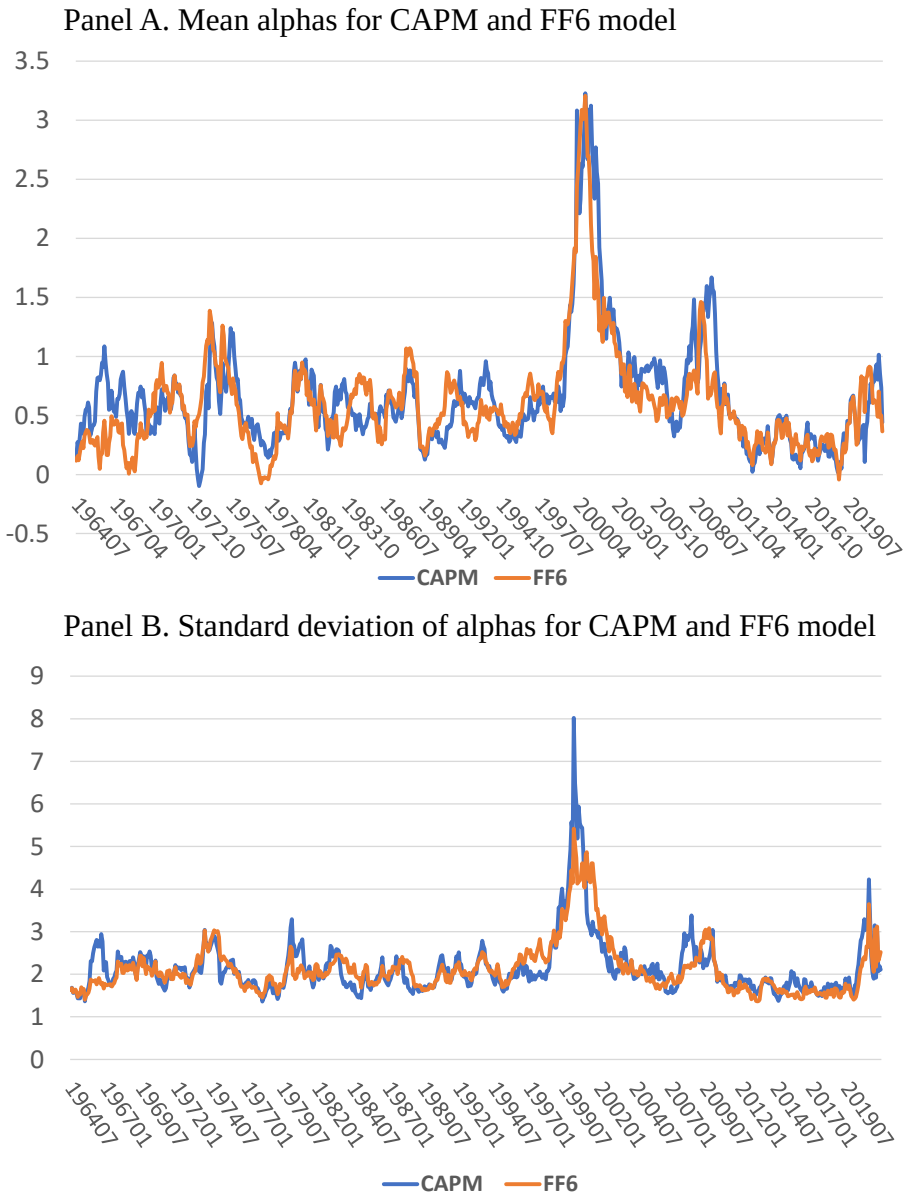
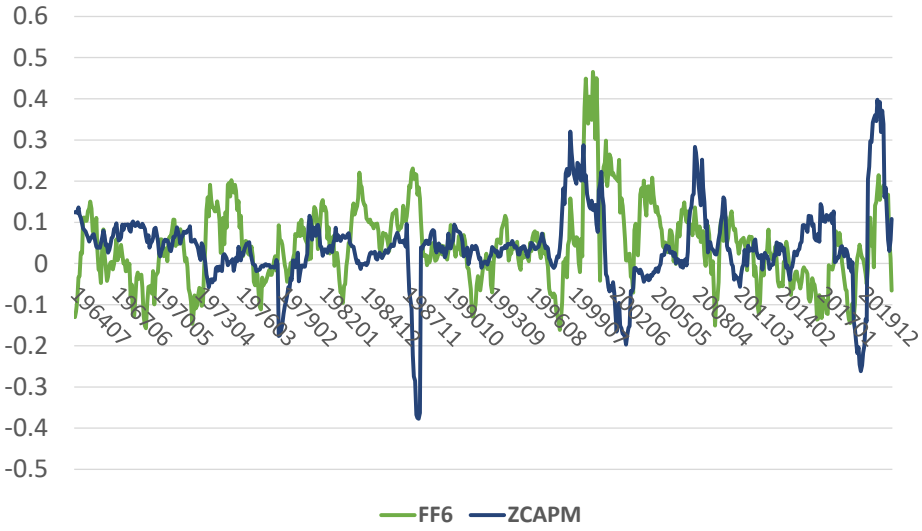


Figure 3: This graph shows one-month rolling mean alpha ($\bar{\alpha}$) values and their standard deviations over time estimated from time-series regressions of different asset pricing models using one year of excess daily returns for the top 1,000 individual stocks with largest market capitalization at year-end of the estimation window. Panel A contains mean alphas for the stocks over time for the CAPM and FF6 model. Panel B plots the standard deviations of alphas for the stocks over time for the CAPM and FF6 model. Both statistics are scaled to monthly percent returns. The time-series regression for each model is initially estimated using daily returns from July 1963 to June 1964, and subsequent regressions are estimated by rolling forward one month at a time until November 2021 to obtain series of 690 values.

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Panel A. Mean alphas for FF6 model and ZCAPM



Panel B. Standard deviations of alphas for FF6 model and ZCAPM

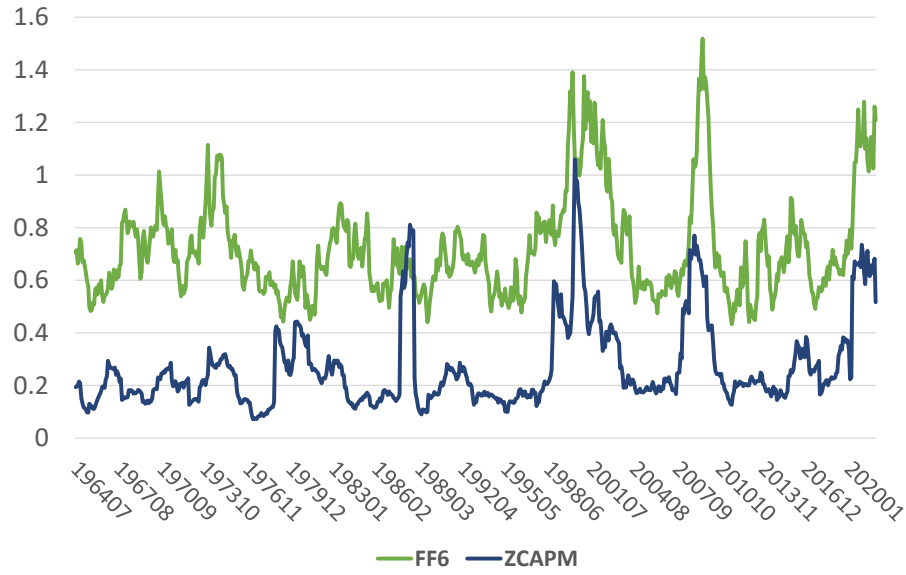


Figure 4: This graph shows one-month rolling mean alpha ($\bar{\alpha}$) values and their standard deviations over time estimated from time-series regressions of the Fama and French six-factor model and ZCAPM using one year of daily excess returns for 222 stock portfolios. Panel A contains mean alphas for the portfolios over time for the FF6 model and ZCAPM. Panel B plots the standard deviations of alphas for the portfolios over time for the FF6 model and ZCAPM. Both statistics are scaled to monthly percent returns. The time-series regression for each model is initially estimated using daily returns from July 1963 to June 1964, and subsequent regressions are estimated by rolling forward one month at a time until November 2021 to obtain series of 690 values.

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Appendix

Based on equation (1) in the text, for an asset pricing model containing a market factor plus K risk factors, a time-series regression can be specified as follows:

$$R_{it}^e = \alpha_i + b_{iM} R_{Mt}^e + \sum_{k=1}^{K} b_{ik} F_{kt} + \varepsilon_{it}, \tag{A.1}$$

where R_{it}^e is the excess return for asset i at time t, R_{Mt}^e is the excess return of market factor, and F_{kt} is the excess return of factor k. In this equation, all betas are dimensionless with no units of measurement, and alphas have the same units of measurement as returns.

Factors can be mimicked by the performance of portfolios consisting of investment assets. The weights for all assets in the portfolios are dimensionless. For the market risk factor mimicked portfolio, we have $\sum_{i=1}^{N} w_i = 1$, where N denotes the total number of assets in the market. For each of the other K risk factors, the weights for their mimicked portfolio satisfy: (1) $\sum_{i=1}^{N} w_i = 0$; and (2) $\sum_{i=1}^{l} w_i^l = 1$ when $w_i^l \geq 0$ (long portfolio weights) and $\sum_{i=1}^{s} w_i^s = -1$ when $w_i^s < 0$ (short portfolio weights) with $l + s = N$. If condition (2) does not hold, then the mimicked portfolio is not well defined; in turn, the factor loading b_{ik} is not well defined. To see this, we can multiply the risk factor mimicked portfolio by an arbitrary constant number, such that the factor loading b_{ik} will shrink by a constant amount that is difficult to interpret.

When estimating an out-of-sample cross-sectional regression model, we have:

$$R_{iT+1}^e = R_z + \eta\alpha_i + \lambda_M b_{iM} + \sum_{k=1}^{K} \lambda_k b_{ik} + u_{iT+1}. \tag{A.2}$$

In this equation, R_z and λ_k coefficients are measured in terms of returns but not η. Following Fama (1976, Chapter 9), we can always find two sets of pure real numbers w_i^l and w_i^s wherein $\sum_{i=1}^{l} w_i^l = 1$ with $w_i^l \geq 0$ (long portfolio) and $\sum_{i=1}^{s} w_i^s = -1$ with $w_i^s < 0$ (short portfolio),

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such that the return on the long/short portfolio is:

$$R_{PT+1} = \sum_{i=1}^{l} w_i^l R_{iT+1} + \sum_{i=1}^{s} w_i^s R_{iT+1} = \lambda_P \tag{A.3}$$

with

$$\begin{aligned}
\sum_{i=1}^{l} w_i^l R_z + \sum_{i=1}^{s} w_i^s R_z &= 0 \\
\sum_{i=1}^{l} w_i^l b_{iM} + \sum_{i=1}^{s} w_i^s b_{iM} &= 1 \\
\sum_{i=1}^{l} w_i^l \alpha_i + \sum_{i=1}^{s} w_i^s \alpha_i &= 0 \\
\sum_{i=1}^{l} w_i^l b_{ik} + \sum_{i=1}^{s} w_i^s b_{ik} &= 0 \text{ for any } k.
\end{aligned} \tag{A.4}$$

We can see that λ_P is the return of a long-short portfolio. Similarly, all λ_k coefficients correspond to the returns of long/short portfolios.

Extending Fama (1976), who did not consider the intercept term (or alpha) in the time-series regression model, we focus on cross-sectional sensitivity coefficient η associated with time-series regression α_i coefficients. Again we can find two sets of pure numbers w_i^ls and w_i^ss wherein $\sum_{i=1}^{l} w_i^l = 1$ with $w_i^l \geq 0$ (long portfolio) and $\sum_{i=1}^{s} w_i^s = -1$ with $w_i^s < 0$ (short portfolio), so that the return on the long/short portfolio is:

$$R_{PT+1} = \sum_{i=1}^{l} w_i^l R_{iT+1} + \sum_{i=1}^{s} w_i^s R_{iT+1} = \eta \left(\sum_{i=1}^{l} w_i^l \alpha_i + \sum_{i=1}^{s} w_i^s \alpha_i \right) \tag{A.5}$$

with

$$\begin{aligned}
\sum_{i=1}^{l} w_i^l R_z + \sum_{i=1}^{s} w_i^s R_z &= 0 \\
\sum_{i=1}^{l} w_i^l b_{iM} + \sum_{i=1}^{s} w_i^s b_{iM} &= 0 \\
\sum_{i=1}^{l} w_i^l b_{ik} + \sum_{i=1}^{s} w_i^s b_{ik} &= 0 \text{ for any } k.
\end{aligned}$$

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In equation (A.5), $\left(\sum_{i=1}^{l} w_i^l \alpha_i + \sum_{i=1}^{s} w_i^s \alpha_i\right)$ is the return of a long-short portfolio. If we set this quantity equal to one, it means that: (1) the return for this portfolio is 100%; and (2) sensitivity coefficient η is dimensionless and simply measures the proportion between the returns of these two portfolios.

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